

# Giseop KIM

## Assistant Professor

Department of Robotics and Mechatronics Engineering, DGIST  
Joint appointments: Department of Artificial Intelligence; Mechanical Engineering Track  
Director, Autonomy and Perceptual Robotics Lab (APRL)

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Google Scholar | GitHub

**Research mission.** APRL advances spatial intelligence and robot world models to enable autonomous robots to navigate, reason, and learn from experience in the real world.

**Research profile.** Robotics researcher working at the intersection of simultaneous localization and mapping (SLAM), spatial and physical AI, 3D perception, sensor fusion, long-term autonomy, and visual-language robot navigation. Current research investigates memory-augmented spatial intelligence, shared robot experience, embodied reasoning, and robot world models for robust real-world navigation.

## Academic Appointments and Professional Experience

|                     |                                                                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Dec. 2024–present   | <b>Assistant Professor</b> , Daegu Gyeongbuk Institute of Science and Technology (DGIST), Daegu, Republic of Korea. Principal Investigator, APRL. |
| Dec. 2021–Dec. 2024 | <b>Research Scientist</b> , NAVER LABS. Research in autonomous driving, 3D vision, mapping, and localization.                                     |
| Mar. 2017–Aug. 2021 | <b>Graduate Researcher</b> , KAIST, Intelligent Robotic Autonomy and Perception Lab.                                                              |

## Education

|      |                                                                                                                                                 |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 | <b>Ph.D., Civil and Environmental Engineering</b> , KAIST. Advisors: <a href="#">Prof. Ayoung Kim</a> and <a href="#">Prof. Youngchul Kim</a> . |
| 2019 | <b>M.S., Civil and Environmental Engineering</b> , KAIST.                                                                                       |
| 2017 | <b>B.S., Civil and Environmental Engineering</b> , KAIST.                                                                                       |

## Research Directions

|                                            |                                                                                                                                                     |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Spatial intelligence for navigation</b> | SLAM 2.0 for the Robot Web era; neural map representations; human-robot interactive visual-language navigation; long-term and multi-robot autonomy. |
| <b>Embodied reasoning and world models</b> | Reasoning for robots; generative AI for mobile robot navigation; memory-augmented spatial intelligence; robot world models.                         |

## Funded Research

PI: Principal Investigator; Co-PI: Co-Principal Investigator; Participant: Participating Researcher.

## Ongoing Projects

| Project / funding details                                                                                                                                                                                                                                                                                                                                                                                       | Role  | Period              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------|
| <b>Robot Industry Technology Development Program</b><br><b>Sponsor:</b> Ministry of Trade, Industry and Energy<br><b>Program:</b> 2026 Robot Industry Technology Development Program (Core Robot Industry Technology Development)<br><b>Title:</b> Spatial-RFM: Development of a VLM-based Spatial Intelligence Robot Foundation Model for Humanoid and Quadruped Robot Navigation in Unstructured Environments | Co-PI | Jul. 2026–Dec. 2029 |
| <b>AIMS: Memory-Augmented Spatial Intelligence</b><br><b>Sponsor:</b> Ministry of Science and ICT<br><b>Program:</b> NRF Early-career Research Program (Type B), Basic Research Program 2026<br><b>Title:</b> Memory-Augmented Spatial Intelligence for Heterogeneous Autonomous Mobility Systems Understanding and Interacting with Implicit Human Instructions                                                | PI    | Mar. 2026–Feb. 2030 |

| Project / funding details                                                                                                                                                                                                                                                                                         | Role        | Period                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------|
| <b>Industry Project 2026-1 (Confidential)</b><br><b>Sponsor:</b> Industry-funded<br><b>Program:</b> Industry-academic technical service<br><b>Title:</b> Mapping and Localization in Factory Environments                                                                                                         | PI          | Apr. 2026–Mar. 2027            |
| <b>Basic Research Laboratory, Exploratory Type</b><br><b>Sponsor:</b> Ministry of Science and ICT<br><b>Program:</b> Basic Research Program, Basic Research Laboratory (Exploratory Type) 2026<br><b>Topic:</b> Fundamental Study on Morphogenetic Cooperative Mobility Intelligence based on Physical Agentic AI | Co-PI       | Jul. 2026–Jun. 2029            |
| <b>Glocal Lab</b><br><b>Sponsor:</b> Ministry of Education<br><b>Program:</b> Glocal Lab Program, Academic Research Support Program for Science and Engineering 2025<br><b>Center:</b> TransHuman Robotics Research Center, Robotics and Mechatronics Research Institute of DGIST                                 | Participant | Sep. 2025–Aug. 2028 (+6 years) |
| <b>InnoCORE</b><br><b>Sponsor:</b> Ministry of Science and ICT<br><b>Title:</b> DGIST Team on Bio-Embodied Physical AI                                                                                                                                                                                            | Participant | Jul. 2025–Dec. 2027 (+3 years) |
| <b>AI Star Fellowship</b><br><b>Sponsor:</b> Ministry of Science and ICT<br><b>Title:</b> Human-centered Symbiotic Embodied AI Systems                                                                                                                                                                            | Participant | 2025–2030                      |
| <b>DGIST Start-up Research</b><br><b>Sponsor:</b> DGIST<br><b>Title:</b> Self-evolving Robot-Web Navigation Intelligence                                                                                                                                                                                          | PI          | Jan. 2025–Dec. 2028            |

## Completed Projects

| Project / funding details                                                                                                                                                                                                                                                                                                                                                 | Role        | Period              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------|
| <b>P2P Autonomous Driving Platform</b><br><b>Sponsor:</b> Ministry of Trade, Industry and Energy<br><b>Program:</b> Autonomous Driving Technology Development Innovation Program, Korea Evaluation Institute of Industrial Technology<br><b>Title:</b> Development of a Level-4 Passenger-Van Autonomous Driving Platform for Point-to-Point Mobility in Designated Areas | Participant | Sep. 2025–Mar. 2026 |
| <b>N-HRHR</b><br><b>Sponsor:</b> DGIST<br><b>Title:</b> Exploring Human-Robot Collaborative Coexistence in the Era of Large-Scale Robots                                                                                                                                                                                                                                  | Participant | Jul.–Dec. 2025      |

## Honors and Awards

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|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sep. 30, 2026 | <b>Best Paper Award</b> , IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026), for “LT-Mem: Volatility-Aware Spatio-Temporal Memory for Life-long Scene Understanding.” Recipients: Yumin Lee, Hyoseok Ju, and Giseop KIM. |
| Jul. 2026     | <b>Outstanding Young Researcher Award</b> , 41st Annual Conference of the Institute of Control, Robotics and Systems (ICROS 2026).                                                                                                                     |
| Feb. 2026     | <b>Early-career Researcher Award</b> , Korea Robotics Society Annual Conference (KRoC 2026).                                                                                                                                                           |
| Oct. 2025     | <b>Best Poster Award</b> , Human-aware Embodied AI Workshop at the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025) (student advisee: Jiseon Kim).                                                                      |

## Academic Leadership and Service

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|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2027 | <b>Workshops &amp; Tutorials Committee Co-Chair</b> , IEEE International Conference on Robotics and Automation (ICRA 2027). <i>Forthcoming</i> .                                                   |
| 2026 | <b>Workshop Organizer</b> , Second Workshop on Human-aware Embodied AI, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026).                                           |
| 2026 | <b>Workshop Organizer</b> , Long-Term Perception for Human-Centric Autonomy: Towards Undo for the Physical World, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026). |
| 2026 | <b>Organizing Committee Member</b> , 41st Annual Conference of the Institute of Control, Robotics and Systems (ICROS 2026).                                                                        |
| 2026 | <b>Tutorial Organizer and Speaker</b> , Spatial AI Tutorial, Korea Robotics Society Annual Conference (KRoC 2026).                                                                                 |

## Publications

### Legend

|                   |                                                             |
|-------------------|-------------------------------------------------------------|
| ►                 | First/co-first or corresponding-author paper by Giseop KIM. |
| *                 | First/co-first author.                                      |
| †                 | Corresponding author.                                       |
| <b>Giseop KIM</b> | The CV author's name is shown in bold.                      |
| "to appear"       | Accepted paper awaiting publication.                        |

### Peer-Reviewed Journal Articles

- J7. S. Lee\*, H. Yu, **Giseop KIM**, and S. Choi†, "LAMP: Implicit Language Map for Robot Navigation," *IEEE Robotics and Automation Letters*, vol. 10, no. 12, pp. 12365–12372, 2025.
- J6. M. Kim\*, **Giseop KIM**, and S. Choi†, "Addressing Diverging Training Costs Using BEVRestore for High-Resolution Bird's Eye View Map Construction," *IEEE Robotics and Automation Letters*, 2024.
- J5. H. Kim\*, J. Choi, T. Sim, **Giseop KIM**, and Y. Cho†, "Narrowing Your FOV with SOLiD: Spatially Organized and Lightweight Global Descriptor for FOV-Constrained LiDAR Place Recognition," *IEEE Robotics and Automation Letters*, 2024.
- J4. M. Jung\*, W. Yang, D. Lee, H. Gil, **Giseop KIM**, and A. Kim†, "HeLiPR: Heterogeneous LiDAR Dataset for Inter-LiDAR Place Recognition under Spatiotemporal Variations," *The International Journal of Robotics Research*, vol. 43, no. 12, pp. 1867–1883, 2024.
- J3. ► **Giseop KIM**\*, S. Choi, and A. Kim†, "Scan Context++: Structural Place Recognition Robust to Rotation and Lateral Variations in Urban Environments," *IEEE Transactions on Robotics*, 2022.
- J2. Y. Cho\*, **Giseop KIM**, S. Lee, and J.-H. Ryu†, "OpenStreetMap-based LiDAR Global Localization in Urban Environment without a Prior LiDAR Map," *IEEE Robotics and Automation Letters*, 2022.
- J1. ► **Giseop KIM**\*, B. Park, and A. Kim†, "1-Day Learning, 1-Year Localization: Long-term LiDAR Localization using Scan Context Image," *IEEE Robotics and Automation Letters*, 2019.

### Peer-Reviewed International Conference Papers

- C19. ► H. Kim\*, B. Kim, and **Giseop KIM**†, "MarsLab: A Martian Rover Simulator for Planetary Rover Autonomous Navigation," *International Conference on Space Robotics (iSpaRo)*, 2026, accepted, to appear.
- C18. ► H. Kim\* and **Giseop KIM**†, "Simulation for Planetary Robotic Perception and Autonomy: A Concise Survey of Recent Capabilities and Gaps," *iSpaRo*, 2026, accepted, to appear.
- C17. ► Y. Lee\*, H. Ju, and **Giseop KIM**†, "LT-Mem: Volatility-Aware Spatio-Temporal Memory for Lifelong Scene Understanding," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. **Best Paper Award**.
- C16. ► B. Nayak\*, H. Ju\*, and **Giseop KIM**†, "Mag4D-SLAM Dataset: A Repeated-Traversal Multi-Modal 4D Geomagnetic Dataset for Localization and Mapping," *IROS*, 2026.
- C15. S. Moon\*, J. Baek, Y. Jeong, D. Chae, **Giseop KIM**, J. Lee, J. Kim†, and S. Choi†, "Streaming Dense Voxel Representations for 3D Occupancy Prediction," *European Conference on Computer Vision (ECCV)*, 2026.
- C14. ► B. Suh\*, H. Ju, and **Giseop KIM**†, "What Matters for Real-World Long-Horizon Robot Navigation?: An Experimental Study of Implicit Goals and Sparse Memory," *23rd IFAC World Congress*, 2026.
- C13. J. Lee\*, J. Kim\*, W. Shin, **Giseop KIM**, and H. Oh†, "Learning Point Cloud Geometry as a Statistical Manifold: Theory and Practice," *Robotics: Science and Systems (RSS)*, 2026.
- C12. ► H. Ju\*, B. Suh, and **Giseop KIM**†, "Have We Mastered Scale in Deep Monocular Visual SLAM? The Scale-Master Dataset and Benchmark," *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- C11. M. Kim\*, D. Lee, J. Yu, J. Hur, **Giseop KIM**, and J. Kim†, "Inlier-Centric Post-Training Quantization for Object Detection Models," *International Conference on Learning Representations (ICLR)*, 2026.

- C10. J. Kim<sup>\*</sup>, S. Jeong<sup>\*</sup>, **Giseop KIM**, M.-H. Jeon, E. Jun, and A. Kim<sup>†</sup>, “2D Gaussian Splatting-based Sparse-view Transparent Object Depth Reconstruction via Physics Simulation for Scene Update,” *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
- C9. H. Gil<sup>\*</sup>, D. Lee<sup>\*</sup>, **Giseop KIM**, and A. Kim<sup>†</sup>, “Ephemerality Meets LiDAR-based Lifelong Mapping,” *ICRA*, 2025.
- C8. M. Kim<sup>\*</sup>, **Giseop KIM**, K.-H. Jin, and S. Choi<sup>†</sup>, “BroadBEV: Collaborative LiDAR-Camera Fusion for Broad-Sighted Bird’s Eye View Map Construction,” *ICRA*, 2024.
- C7. H. Lim<sup>\*</sup>, K. Han, G. Shin, **Giseop KIM**, S. Hong, and H. Myung<sup>†</sup>, “ORORA: Outlier-Robust Radar Odometry,” *ICRA*, 2023.
- C6. S. Yun<sup>\*</sup>, M. Jung, J. Kim, S. Jung, Y. Cho, M.-H. Jeon, **Giseop KIM**, and A. Kim<sup>†</sup>, “STheReO: Stereo Thermal Dataset for Research in Odometry and Mapping,” *IROS*, 2022.
- C5. ► **Giseop KIM**<sup>\*</sup> and A. Kim<sup>†</sup>, “LT-mapper: A Modular Framework for LiDAR-based Lifelong Mapping,” *ICRA*, 2022.
- C4. ► **Giseop KIM**<sup>\*</sup> and A. Kim<sup>†</sup>, “Remove, then Revert: Static Point Cloud Map Construction using Multiresolution Range Images,” *IROS*, 2020.
- C3. ► **Giseop KIM**<sup>\*</sup>, Y. S. Park, Y. Cho, J. Jeong, and A. Kim<sup>†</sup>, “MulRan: Multimodal Range Dataset for Urban Place Recognition,” *ICRA*, 2020.
- C2. Y. Cho<sup>\*</sup>, **Giseop KIM**, and A. Kim<sup>†</sup>, “Unsupervised Geometry-aware Deep LiDAR Odometry,” *ICRA*, 2020.
- C1. ► **Giseop KIM**<sup>\*</sup> and A. Kim<sup>†</sup>, “Scan Context: Egocentric Spatial Descriptor for Place Recognition within 3D Point Cloud Map,” *IROS*, 2018.

### Book Chapters

- B1. J. Behley, M. Fallon, S. Zhao, **Giseop KIM**, J. Zhang, F. Zhang, and A. Kim, “LiDAR SLAM,” Chapter 8 in *SLAM Handbook: From Localization and Mapping to Spatial Intelligence*, 2026.

### Domestic Journal Articles

- DJ1. ► B. Suh<sup>\*</sup>, J. Kim, J. Son, M. Park, and **Giseop KIM**<sup>†</sup>, “Knowledge Fusion Strategies for Object Goal Navigation in the VLM Era: A Survey,” *Transactions of the Korean Society of Automotive Engineers*, vol. 34, no. 1, pp. 119–131, 2026.

### Domestic Conference Papers

- DC3. ► A. Lee<sup>\*</sup> and **Giseop KIM**<sup>†</sup>, “Floorplan-based Point-to-Language Grounded Navigation,” Institute of Control, Robotics and Systems Annual Conference (ICROS), 2026.
- DC2. ► A. Lee<sup>\*</sup>, H. Yoon<sup>\*</sup>, S. Park<sup>\*</sup>, M. Lee<sup>\*</sup>, and **Giseop KIM**<sup>†</sup>, “Rendezvous Briefing: VLM-based Semantic-Aware Multi-Robot Collaborative Exploration,” ICROS, 2026.
- DC1. ► **Giseop KIM**<sup>\*†</sup>, M. Park, and J. Son, “Change-Robust LiDAR Localization in Urban Environments,” Institute of Electronics and Information Engineers Conference, 2025.

### Invited Talks and Tutorials

#### Conference

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|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sep. 27, 2026 | “From Maps to Memory: Bridging the Perception–Reasoning Gap for Long-Term Robot Navigation,” Workshop on Bridging Navigation, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026). |
| Jul. 3, 2026  | “Visual Language Multi-robot SLAM,” Early-career Researcher Award Session, 41st Annual Conference of the Institute of Control, Robotics and Systems (ICROS 2026), Daegu.                                       |
| Feb. 6, 2026  | “IMU Basics and Inertial Aided Navigation,” Spatial AI Tutorial Organizer and Speaker, Korea Robotics Society Annual Conference (KRoc 2026), Pyeongchang.                                                      |
| Feb. 5, 2026  | “Memory-Augmented Spatial Intelligence for Autonomous Robot Navigation,” Early-career Researcher Award Session, KRoc 2026, Pyeongchang.                                                                        |
| Feb. 5, 2026  | “CoT, Scene Graphs, and Recent Paradigms for Mobile Robot Navigation: A Short Survey and Real-World Lessons,” Special Session on AI-based Autonomous Systems, KRoc 2026, Pyeongchang.                          |
| Jan. 19, 2026 | “An Invitation to 3D Vision, Spatial AI, and Physical AI,” Invited Tutorial, International Conference on Electronics, Information, and Communication (ICEIC 2026), Macau.                                      |
| Oct. 30, 2025 | “Physical AI for Smart Factory: Humanoid, Sensing, and Spatial Intelligence,” Fall Conference of the Korean Fiber Society, Busan.                                                                              |
| Aug. 15, 2025 | “Generative AI for Mobile Robot Navigation,” AI Robotics KR Conference: Human and AI Robotics Together, Seoul.                                                                                                 |
| Jul. 18, 2025 | “Generative AI for Mobile Robot Navigation,” Joint Summer School on Image Understanding and Image Processing, Institute of Electronics and Information Engineers, Seoul.                                       |

## University

- Jul. 24, 2026 “Bridging the Perception–Reasoning Gap for Long-Term Robot Navigation (Feat. the Research Journey),” Invited Talk, Korea Advanced Institute of Science and Technology (KAIST).
- May 28, 2026 “3D Perception, Language, and AI for Human-Robot Interactive Navigation,” Ewha Womans University.
- Apr. 2, 2026 “Memory-Augmented Spatial Intelligence for Autonomous Robot Navigation,” Hanyang University ERICA.
- Dec. 10, 2025 “SLAM, Spatial AI, and Human-Robot Interactive Navigation,” Department of Intelligent Robotics, Sungkyunkwan University, Suwon.
- Dec. 9, 2025 “Robotics and Spatial AI for Construction,” Department of Civil and Environmental Engineering, KAIST, Daejeon.

## Research Institute

- Sep. 4, 2026 “Lidar Inertial SLAM for Spaceship Docking,” Invited Seminar, Korea Aerospace Research Institute (KARI).
- Aug. 19, 2026 “From Maps to Memory: Bridging Perception and Reasoning for Long-Term Autonomous Navigation,” Invited Seminar, Korea Institute for Robot Industry Advancement (KIRIA), Seoul.
- Jul. 30, 2026 “Geometric 3D Vision Made Easy? – The DUS3R Family and Its Evolution,” Invited Seminar, Electronics and Telecommunications Research Institute (ETRI), Daegu-Gyeongbuk Regional Research Division.
- Mar. 24, 2026 “Memory-Augmented Spatial Intelligence for Autonomous Robot Navigation,” ETRI Daegu-Gyeongbuk Regional Research Division.
- Dec. 8, 2025 “SLAM, Spatial AI, and Human-Robot Interactive Navigation,” Department of Industrial Machinery DX, Korea Institute of Machinery and Materials, Daejeon.
- May 28, 2025 “Spatial AI, from the 2000s to 2025: The Evolution of Spatial Intelligence and the Future of Robotics,” ETRI Daegu-Gyeongbuk Regional Research Division Mid- to Long-term Research Roadmap Meeting, Daegu.

## Industry

- Jun. 2, 2026 “What, When, and Why Visual-Language Navigation?,” Private Meetup at the IEEE International Conference on Robotics and Automation (ICRA 2026), URobotics.
- Jun. 13, 2025 “Lessons from Translating Research into Products and Services,” SLAM and Perception Seminar, Hyundai Motor Company Uiwang Research Institute.

## Public Sector

- Feb. 2, 2026 “Daegu’s Innovation, Moving Towards a Robot and Future Mobility City,” Dalseo-gu Office Monthly Meeting Special Lecture, Daegu.

## Graduate Student Advising

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|------------|-----------------------|---------------------|
| Bokeon Suh | Integrated M.S./Ph.D. | Fall 2025–present   |
| Jiseon Kim | M.S.                  | Fall 2025–present   |
| Yumin Lee  | M.S.                  | Fall 2025–present   |
| Hyoseok Ju | Integrated M.S./Ph.D. | Fall 2025–present   |
| Doyeon Kim | Ph.D.                 | Spring 2026–present |
| Beomsu Kim | M.S.                  | Spring 2026–present |
| Hoyun Kim  | M.S.                  | Spring 2026–present |

## Teaching at DGIST

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|             |         |                                         |
|-------------|---------|-----------------------------------------|
| 2026 Fall   | RT616   | <a href="#">Vision/Image Processing</a> |
| 2026 Fall   | TM563   | AI-based Autonomous Robot Systems       |
| 2026 Spring | RT604   | Advanced Mobile System                  |
| 2026 Spring | MECH301 | Robots for Human                        |
| 2025 Fall   | MECH307 | Introduction to Artificial Intelligence |
| 2025 Fall   | AT603   | Introduction to Mobility Engineering    |
| 2025 Spring | BE203   | Creative Mechanical Design              |

## Research Group

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**Autonomy and Perceptual Robotics Lab (APRL), DGIST.** Founded December 2024. Current full-time group: one postdoctoral researcher, three doctoral researchers (including two integrated M.S./Ph.D. students), and four M.S. students, with undergraduate interns and external collaborators across robotics, perception, and AI.